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Project title: Logistics Performance & Customer Satisfaction Analysis

Grouping

I work on the project individually.

Domain Area

Project Focus: **E-commerce Marketplace Operations**

Industry & Sector: This project addresses the Retail industry within the Digital Commerce (E-commerce) sector. It specifically examines the Marketplace model, where a central platform (Olist) acts as an intermediary connecting small and medium-sized Brazilian businesses with customers across various online sales channels.

Encompassed Knowledge: The domain encompasses several key business functions, including:

Logistics & Supply Chain: Tracking the journey from order placement to final delivery, including freight performance and shipping lead times.

Customer Experience: Analyzing satisfaction through review scores and textual feedback.

Financial Transactions: Managing diverse payment methods (e.g., credit cards, Boletos) and revenue distribution across product categories.

Geospatial Analysis: Understanding regional market penetration across Brazil's 27 states using geolocation data.

General Boundaries: The study is bounded by real-world commercial data from Olist Store spanning from September 2016 to September 2018. It covers approximately 100,000 orders involving thousands of unique products and sellers, providing a comprehensive view of the Brazilian e-commerce landscape during this period.

Dataset

Name of the source of the dataset:

Brazilian E-Commerce Public Dataset by Olist

Website link to the original dataset (uncleaned):

<https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce>

Data Cleaning

Environment: Python (Anaconda / Jupyter Notebook)

Primary Libraries: Pandas, NumPy

Exploratory Data Analysis (EDA):

`df.info()` : check data type and non-null counts

`df.isnull().sum()` : check missing value statistics

`df.describe(include = 'all')` : check statistical summary of both numerical and categorical data.

1. Customer Dataset [**olist_customers_dataset.csv**]

Zip Code Integrity: Convert **customer_zip_code_prefix** from numeric to **String** type, remove any floating-point decimals (e.g., .0), and apply `.str.zfill(5)` to ensure a fixed 5-digit format.

Reason: Prevents dropping leading zeros (e.g., 01001 becoming 1001).

2. Geolocation Dataset [**olist_geolocation_dataset.csv**]

Zip Code Integrity: Convert **geolocation_zip_code_prefix** from numeric to **String** type, remove any floating-point decimals (e.g., .0), and apply `.str.zfill(5)` to ensure a fixed 5-digit format.

Reason: Prevents dropping leading zeros (e.g., 01001 becoming 1001), ensuring valid Brazilian geographic mapping.

Deduplication: Remove 261,831 duplicate rows.

Outlier Removal: Filter coordinates to remain within Brazil's boundaries (removes 33 error records).

Benefit: Ensures Power BI maps focus accurately on Brazil without global distortion.

Coordinate Aggregation (Handling Duplicates)

A single zip code prefix often has multiple lat/lng entries in this file (e.g., 01001 appears multiple times with slightly different coordinates).

Action: Group by **geolocation_zip_code_prefix** and calculate the Mean (Average) of the Latitude and Longitude for each unique zip code.

Reasoning: Reduces the dataset size significantly (from ~0.74 million rows (after deduplication) to ~19,000 unique prefixes), improving Power BI performance while maintaining a highly accurate "center point" for each postal area.

3. Order Items Dataset [**olist_order_items_dataset.csv**]

DataType Optimization: Keep **order_item_id** as Integer.

Reason: Enables sequential calculations (e.g., identifying bulk orders where **item_id** > 5 or max items per order).

Temporal Conversion: Convert **shipping_limit_date** to Datetime to calculate vendor shipping lead times.

4. Payments Dataset [**olist_order_payments_dataset.csv**]

Financial Accuracy: Remove records where **payment_installments** is 0, **payment_value** is 0, or **payment_type** is 'not_defined'. [11 rows removed]

Reason: Eliminates non-economic noise that skews Average Order Value (AOV) and revenue KPIs.

5. Order Reviews Dataset [**olist_order_reviews_dataset.csv**]

Data Enrichment: Fill missing titles/messages with "No Title" and "No Message."

Datetime Normalization: Convert all review timestamps to Datetime.

Review Logic: Keep only the latest review per **order_id** to prevent redundant scoring bias in satisfaction analysis.

6. Orders Dataset [**olist_orders_dataset.csv**]

Missing Date Policy: Preservation of NaT/Null Values

Maintain all missing values as NaT (Not-a-Time) in Python/Pandas and Null/Empty in Power BI.

Critical Reason: Prevents "1970 Epoch" errors and avoids breaking Power BI Date Slicers. Keeping values empty preserves native Datetime functionality, ensuring that columns are not forced into "**String**" types, which would disable time-series calculations and DAX date functions.

Business Logic Interpretation: Missing timestamps are treated as functional status indicators of the order lifecycle rather than data entry errors:

order_approved_at (160 nulls): Orders where payment was never confirmed or was canceled post-creation.

order_delivered_carrier_date (1,783 nulls): Orders that were never picked up by the logistics partner.

order_delivered_customer_date (2,965 nulls): Active "In-Transit" orders, lost shipments, or canceled orders.

Analytical Strategy: Context-Specific Filtering

Instead of permanent deletion from the master dataset, records will be filtered dynamically based on the specific metric being analyzed:

Sample Case: Shipping Duration Analysis

Metric: Actual Shipping Days (**order_delivered_customer_date - order_purchase_timestamp**).

Action: Exclude records where **order_delivered_customer_date** is Null.

Purpose: This ensures that "Actual Shipping Days" reflects only successfully completed shipments, preventing calculation errors (NaN) or skewed results from canceled or pending orders.

7. Products Dataset [**olist_products_dataset.csv**]

Imputation Strategy:

Fill missing photo, title length and description length counts with 0.

Fill physical dimensions (weight/length/height/width) with the Median to avoid distribution bias.

8. Seller Dataset [**olist_sellers_dataset.csv**]

Zip Code Integrity: Convert **seller_zip_code_prefix** from numeric to **String** type, remove any floating-point decimals (e.g., .0), and apply `.str.zfill(5)` to ensure a fixed 5-digit format.

Reason: Prevents dropping leading zeros (e.g., 01001 becoming 1001).

Data Filtering: Identified and **removed one record** where the **seller_city** was erroneously entered as a numeric string ('04482255').

Reason: To maintain the integrity of categorical geographical data by removing non-alphabetic anomalies.

9. Translation Dataset [**product_category_name_translation.csv**]

Label missing categories as '**outro**' (Other).

Translation Mapping: Use concat to manually append the 3 missing category translations to the mapping table for 100% reporting coverage.

Model Optimization: Resolving Relationship Ambiguity and Data Redundancy

Problem Statement:

Connecting a single Geolocation table to both the Seller and Customer tables via active 1:M relationships creates a relationship loop (circular path) within the schema. Such loops must be avoided because they generate ambiguous data paths, preventing the Power BI engine from determining a unique filter direction. This ambiguity typically forces the engine to mark one relationship as inactive, resulting in unreliable reports where filters fail to propagate correctly across the model.

Proposed Solution:

I have decoupled these paths by creating two distinct dimension tables: **'Geolocation for Seller'** and **'Geolocation for Customer'**. This approach eliminates ambiguity, allows for independent geographical analysis, and ensures consistent data behavior.

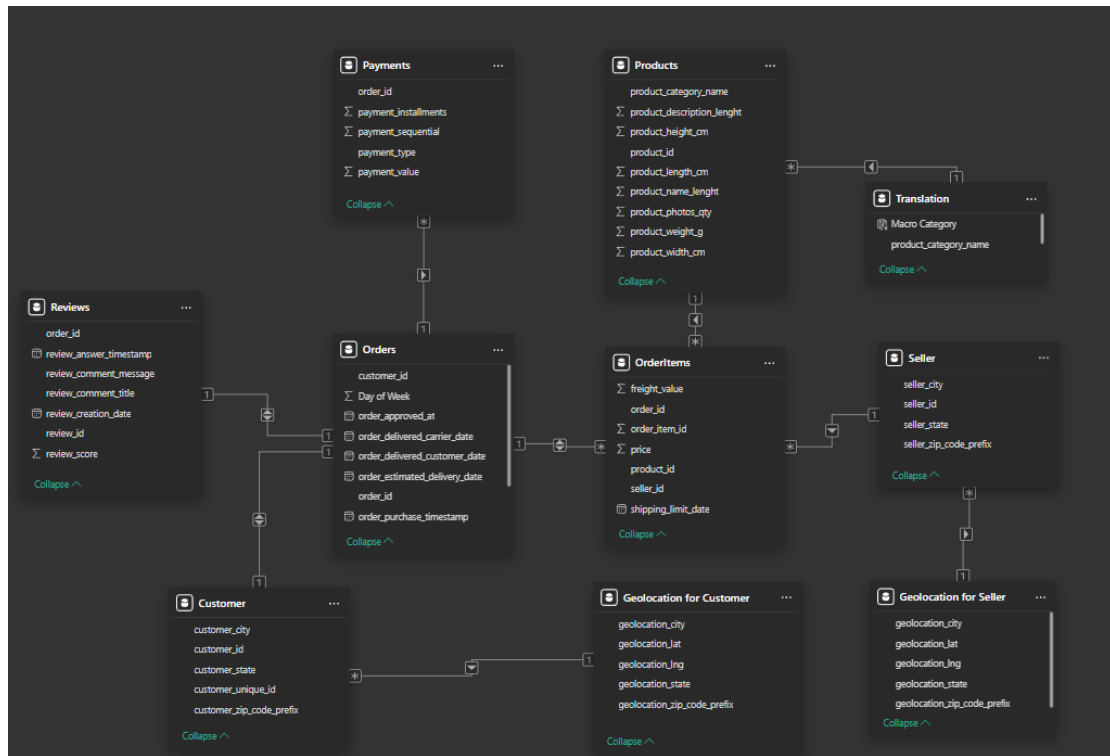
'Geolocation for Seller': Filtered via an Inner Join between **geolocation_zip_code_prefix** and **seller_zip_code_prefix** from the **Seller** table. This restricts the table to only include locations where active sellers exist.

'Geolocation for Customer': Filtered via an Inner Join between **geolocation_zip_code_prefix** and **customer_zip_code_prefix** from the **Customer** table. This ensures the table only contains coordinates relevant to the customer base.

Expected Outcome:

This optimization results in a streamlined data model with a reduced memory footprint, faster query response times, and a "clean" user interface where geographical filters behave predictably for both business perspectives.

Finalized Entity-Relationship Diagram



Note: Some columns in the **Entity-Relationship Diagram** are **created columns**; these columns **will not be** included in the **dataset descriptions**.

Dataset Description for Cleaned Dataset

1st Table Name: **[Customer]**

Purpose: This dataset provides essential geographic and identity information for the buyer. It is used to map delivery destinations and distinguish between individual transactions and unique long-term customers.

Structure: The file contains 99441 rows and 5 columns. Each row represents a single transaction.

Key Variables:

- **customer_id:** The **Primary Key** of this dataset. It contains 99,441 unique values, establishing a **1:1 relationship** with the Orders dataset. A unique customer_id is generated for every order; therefore, the same physical customer will have a different customer_id for each transaction they make.
- **customer_unique_id:** Unique identifier of a customer (i.e. same customer must have same customer_unique_id).
- **customer_zip_code_prefix:** The first five digits of the customer's zip code.
- **customer_city:** The name of the city where the customer is located.
- **customer_state:** The Brazilian state code of customer (e.g., SP, RJ).

Relationship to Other Tables:

This table is linked to **[Orders]** via the **customer_id** field.
(1:1 relationship)

This table is linked to **[Geolocation for Customer]** via the **customer_zip_code_prefix** and **geolocation_zip_code_prefix** in **[Geolocation for Customer]**. **(M:1 relationship)**

2nd Table Name: **[Geolocation for Customer]**

Purpose: This dataset has information Brazilian zip codes and its lat/lng coordinates for customer. Use it to plot maps and find distances between sellers and customers.

Structure: The file contains 14836 rows and 5 columns. Each row represents a single transaction.

Key Variables:

- **geolocation_zip_code_prefix**: The first five digits of the Brazilian postal code. This serves as the linking key to both the Customer and Seller datasets.
- **geolocation_lat**: The precise latitude coordinate of the location.
- **geolocation_lng**: The precise longitude coordinate of the location.
- **geolocation_city**: The name of the city associated with the coordinates.
- **geolocation_state**: The two-letter Brazilian state code.

Relationship to Other Tables: This file is linked to **[Customer]** via the **geolocation_zip_code_prefix** and **customer_zip_code_prefix** from **[Customer]**. (1:M relationship)

3rd Table Name: **[OrderItems]**

Purpose: This dataset provides the geographic coordinates (latitude and longitude) for Brazilian zip codes. It is instrumental for spatial analysis, allowing for the visualization of order distributions on maps and the calculation of shipping distances between sellers and customers.

Structure: The file contains 112650 rows and 7 columns. Each row represents a single transaction.

Key Variables:

- **order_id**: The unique identifier for each order. It acts as a **Foreign Key** to link this dataset with the **Orders Dataset**.
- **order_item_id**: A sequential number identifying the items included in the same order (e.g., if an order has 3 items, the values will be 1, 2, and 3).
- **product_id**: The unique identifier for the product. It serves as a link to the **Products Dataset**.
- **seller_id**: The unique identifier for the seller. It serves as a link to the **Sellers Dataset**.
- **shipping_limit_date**: The deadline for the seller to hand over the order to the logistics partner.
- **price**: The unit price of the item.
- **freight_value**: The shipping cost for the item. (Note: For multi-item orders, the total freight is distributed among the items).

Relationship to Other Tables:

This table is linked to **[Seller]** through the **seller_id** field.

(M:1 relationship)

This table is linked to **[Orders]** through the **order_id** field.

(M:1 relationship)

This file is linked to **[Products]** through the **product_id** field.

(M:1 relationship)

4th Table Name: **[Payments]**

Purpose: This dataset provides details on the payment methods and financial structures of each order. It is essential for understanding customer purchasing power and ensuring the financial accuracy of revenue KPIs.

Structure: The file contains 103875 rows and 5 columns. Each row represents a single transaction.

Key Variables:

- **order_id**: The unique identifier for each order. It serves as a **Foreign Key** to link with the **Orders Dataset**.
- **payment_sequential**: A sequence number identifying multiple payment methods used for a single order (e.g., a customer paying with two different credit cards)
- **payment_type**: The method of payment chosen (e.g., credit card, boleto, voucher).
- **payment_installments**: The number of installments selected by the customer for the transaction.
- **payment_value**: The monetary value of the transaction.

Relationship to Other Tables: This table is linked to **[Orders]**

through the **order_id** field. **(M:1 relationship)**

5th Table Name: **[Reviews]**

Purpose: This dataset captures customer feedback and sentiment regarding the shopping and fulfillment experience. After the estimated or actual delivery, customers receive a survey to rate their satisfaction. This data is critical for quantifying how operational performance (speed/accuracy) translates into brand reputation.

Structure: The file contains 98673 rows and 7 columns. Each row represents a single transaction.

Key Variables:

- **review_id**: The unique identifier for the review content. Note that the same review_id can occasionally be mapped to multiple orders.
- **order_id**: The **Foreign Key** used to link feedback to specific transactions in the Orders Dataset.
- **review_score**: A satisfaction rating ranging from **1 to 5**.
- **review_comment_title**: The title of the review (in Portuguese).
- **review_comment_message**: Detailed feedback comments (in Portuguese).
- **review_creation_date**: The date the survey was sent to the customer.
- **review_answer_timestamp**: The actual timestamp when the customer responded to the survey.

Relationship to Other Tables: This table is linked to **[Orders]** through the **order_id** field. (**1:1 relationship**)

6th Table Name: **[Orders]**

Purpose: This is the core dataset of the project, serving as the central hub that connects customers, items, payments, and reviews. It tracks the complete lifecycle of an order from the initial purchase to final delivery, enabling the calculation of logistics performance and delivery precision.

Structure: The file contains 99441 rows and 8 columns. Each row represents a single transaction.

Key Variables:

- **order_id**: The **Primary Key** for this dataset. It links to items, payments, and reviews.
- **customer_id**: A **Foreign Key** that links to the **Customer Dataset**. (Establishing a 1:1 relationship for each specific transaction).
- **order_status**: The current state of the order (e.g., delivered, shipped).
- **order_purchase_timestamp**: The exact time the customer placed the order.
- **order_approved_at**: The timestamp when the payment was validated.
- **order_delivered_carrier_date**: The timestamp when the order was handed over to the logistics partner.
- **order_delivered_customer_date**: The actual date the product reached the customer.
- **order_estimated_delivery**: The estimated arrival date promised to the customer at the time of purchase.

Relationship to Other Tables:

This table is linked to **[Reviews]** through the **order_id** field. **(1:1 relationship)**

This table is linked to **[Payments]** through the **order_id** field. **(1:M relationship)**

This table is linked to **[OrderItems]** through the **order_id** field.
(1:M relationship)

This table is linked to **[Customers]** through the **customer_id** field.
(1:1 relationship)

7th Table Name: **[Products]**

Purpose: This dataset provides detailed physical attributes and metadata for each product sold on the platform.

Structure: The file contains 32951 rows and 9 columns. Each row represents a single transaction.

Key Variables:

- **product_id**: The **Primary Key** for this dataset, uniquely identifying each product
- **product_category_name**: The root category of the product (in Portuguese).
- **product_name_length**: The number of characters extracted from the product name.
- **product_description_length**: The number of characters extracted from the product description.
- **product_photos_qty**: The number of published photos for the product.
- **product_weight_g**: The product weight measured in grams.
- **product_length_cm**: The product length measured in centimeters.
- **product_height_cm**: The product height measured in centimeters.
- **product_width_cm**: The product width measured in centimeters.

Relationship to Other Tables:

This table is linked to **[OrderItems]** via the **product_id** field.
(1:M relationship)

This table is linked to **[Translation]** via the **product_category_name** field. **(M:1 relationship)**

8th Table Name: **[Seller]**

Purpose: This dataset provides essential information about the sellers who fulfill orders on the Olist platform. It is primarily used to identify seller locations and determine the origin point of the fulfillment process, enabling the calculation of shipping distances and regional seller performance.

Structure: The file contains 3094 rows and 4 columns. Each row represents a single transaction.

Key Variables:

- **seller_id:** The **Primary Key** for this dataset, uniquely identifying each merchant.
- **seller_zip_code_prefix:** The first five digits of the seller's postal code.
- **seller_city:** The name of the city where the seller is based.
- **seller_state:** : The two-letter Brazilian state code (e.g., SP, PR) of the seller.

Relationship to Other Tables:

This table is linked to **[OrderItems]** via the **seller_id** field. **(1:M relationship)**

This table is linked to **[Geolocation for Seller]** via the **seller_zip_code_prefix** and **geolocation_zip_code_prefix** in **[Geolocation for Seller]**. **(M:1 relationship)**

9th Table Name: **[Translation]**

Purpose: This dataset includes data about product category and translated product category.

Structure: The file contains 74 rows and 2 columns. Each row represents a single transaction.

Key Variables:

- **product_category_name:** The original category name in Portuguese, acting as the **Foreign Key** to the **Products Dataset**.
- **product_category_name_english:** The corresponding category name translated into English.

Relationship to Other Tables:

This table is linked to **[Products]** through the field **product_category_name**. **(1:M relationship)**

10th Table Name: **[Geolocation for Seller]**

Purpose: This dataset has information Brazilian zip codes and its lat/lng coordinates for seller. Use it to plot maps and find distances between sellers and customers.

Structure: The file contains 2239 rows and 5 columns. Each row represents a single transaction.

Key Variables:

- **geolocation_zip_code_prefix**: The first five digits of the Brazilian postal code. This serves as the linking key to both the Customer and Seller datasets.
- **geolocation_lat**: The precise latitude coordinate of the location.
- **geolocation_lng**: The precise longitude coordinate of the location.
- **geolocation_city**: The name of the city associated with the coordinates.
- **geolocation_state**: The two-letter Brazilian state code.

Relationship to Other Tables: This file is linked to **[Customer]** via the **geolocation_zip_code_prefix** and **customer_zip_code_prefix** from **[Customer]**. (1:M relationship)

Timetable

Revised Project Timetable

Date	Activities & Deliverables	Responsible	Notes
08/03	Submit Deliverable 1	Tim	Final Deadline
15/03	Complete Deliverable 2 Development	Tim	Internal Milestone
22/03	Submit Deliverable 2	Tim	Final Deadline (Data modeling & visuals)
05/04	Power BI Dashboard Refinement	Tim	UI/UX & Layout optimization
12/04	Finalize Deliverable 3 Draft	Tim	Internal Review
19/04	Submit Deliverable 3 + Presentation	Tim	Final Deadline
26/04	Complete Diagnostic & Predictive Analysis	Tim	Data analysis phase for D4
06/05	Finalize Deliverable 4 Draft	Tim	Internal Review
10/05	Submit Deliverable 4 + Presentation	Tim	Final Deadline